

Slowing of Electroencephalogram in Rapid Eye Movement Sleep Behavior Disorder

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Rapid eye movement (REM) sleep behavior disorder (RBD) is characterized by a loss of atonia and an increase in phasic muscle activity during REM sleep, leading to complex nocturnal motor behaviors. Brainstem structures responsible for the pathogenesis of RBD are also implicated in cortical activation. To verify the hypothesis that electroencephalogram (EEG) activation will be impaired in RBD, we performed quantitative analyses of waking and REM sleep EEG in 15 idiopathic RBD patients and 15 age- and gender-matched healthy subjects. During wakefulness, RBD patients showed a considerably higher θ power in frontal, temporal, and occipital regions with a lower β power in the occipital region. The dominant occipital frequency was significantly lower in RBD. During REM sleep, β power in the occipital region was lower in RBD. This study shows for the first time an impaired cortical activation during both wakefulness and REM sleep in idiopathic RBD, despite an absence of changes on sleep architecture compared with controls. EEG slowing in these patients may represent an early sign of central nervous system dysfunction, perhaps paralleled by subclinical cognitive deficits. The topographical distribution of EEG slowing and possible pathophysiological mechanisms are discussed in light of the known association between RBD and neurodegenerative disorders.

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Rapid eye movement (REM) sleep behavior disorder (RBD) is a condition characterized by a loss of REM sleep muscle atonia and an increase of phasic muscular activity during REM sleep.¹ Patients exhibit complex and potentially harmful nocturnal motor behaviors, usually associated with dream mentation, that can result in injuries to themselves or their bedpartner.² RBD affects mainly older men and may occur alone (idiopathic form) or in association with a variety of neurological disorders.²⁻⁵ In particular, RBD has been found to be strongly associated with a category of neurodegenerative diseases called *synucleinopathies*⁶ that include Parkinson's disease (PD),⁷⁻¹⁰ dementia with Lewy bodies (DLB),¹¹⁻¹⁴ and multiple system atrophy.¹⁵⁻¹⁷

The pathogenesis of RBD is unclear. Evidences of an involvement of several brainstem structures mainly located into the pons, which include the ventral mesopontine junction, the pedunculopontine nucleus (PPN), the laterodorsal tegmental nucleus, the locus coeruleus (LC), and the peri-LC area, come from animal models^{18,19} and from neuropathological observa-

tions in humans.²⁰⁻²³ Recently, it has been suggested that a dopaminergic striatal impairment may be implicated in the pathogenesis of RBD.^{9,24} Indeed, brain imaging studies have shown a decreased nigrostriatal dopaminergic innervation²⁵ and a reduced presynaptic DA transporter binding in the striatum²⁶ in patients with idiopathic RBD. Some of these structures, namely, the PPN region and the LC, which represent also the largest clusters of cholinergic and noradrenergic neurons within the brainstem reticular formation, respectively,²⁷ are known to play a role in arousal, cortical activation, and cognitive functions.²⁸⁻³² A modulatory role of the dopaminergic system in behavioral arousal and alertness also has been documented.^{27,33-35}

To date, no study has assessed the cortical activity in RBD patients. The aim of this investigation was to perform a quantitative analysis of EEG during wakefulness and REM sleep in idiopathic RBD patients, to test the hypothesis that cortical activation will be impaired, as a result of a dysfunction of subcortical activating systems.

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Subjects and Methods

Subjects

Fifteen patients (10 men and 5 women; mean age, 66.2 ± 5.0 years) with polysomnographically confirmed idiopathic RBD and 15 age- and gender-matched healthy control subjects without any clinical evidence of sleep disorder (10 men and 5 women; mean age, 65.7 ± 6.6 years) were studied. Patients had a mean duration of clinical symptoms of 7.3 ± 4.2 years. They underwent a standard neurological examination before the polysomnographic (PSG) recording to rule out the presence of neurological abnormalities including signs of neurodegenerative disease. In particular, RBD patients received specific measures of extrapyramidal dysfunction, as the motor scale (section III) of the Unified Parkinson's Disease Rating Scale,³⁶ to exclude concomitant signs of parkinsonism (mean score, 2 ± 2.8). Seven of the patients had a neuroimaging study (2 computed tomography scan and 5 magnetic resonance imaging) which was normal in four cases and showed nonspecific lesions such as a single lacunar lesion in the white matter (in two cases) and a mild cortical atrophy (in one case). In both patients and control subjects, the presence of affective disorders or dementia was ruled out according to the corresponding Diagnostic and Statistical Manual IV criteria.³⁷ A history of significant head trauma, seizures, encephalopathy of all causes, cerebrovascular disease, severe diabetes, and the presence of sleep disorders such obstructive sleep apnea syndrome, narcolepsy, or restless legs syndrome also were exclusion criteria. The presence of periodic leg movements during sleep (PLMS) did not constitute an exclusion criterion because an excessive number of PLMS are observed in most patients with idiopathic RBD.³⁸ Twelve of 15 RBD patients had a PLMS index greater than 10, and the PLMS index for the whole group was 26.7 ± 15.8 . Both the RBD patients and control subjects were free from psychotropic medications for at least 2 weeks before the recording. The protocol was approved by the hospital ethics committee and subjects gave written informed consent.

Electroencephalogram Recordings and Scoring

Subjects underwent one night of PSG recording in the sleep laboratory. Sleep was recorded and scored according to a modified version of Rechtschaffen and Kales' method.³⁹ In fact, for both groups, REM sleep was scored without the chin electromyogram (EMG) criterion, allowing the presence of muscular tone during REM sleep. The percentage of REM sleep atonia and that of phasic EMG activity has nonetheless been calculated. All REM sleep measures have been performed according to a method described elsewhere.^{10,38,40} Several PSG variables were measured, namely, sleep latency, total sleep time, sleep efficacy, and percentage of stages 1 and 2, slow-wave sleep, and REM sleep. Sleep efficacy represented the percentage of time spent asleep over the total recording time from sleep onset to the last awakening. Beside the standard leads recommended to evaluate sleep stages (left and right electrooculograms, chin EMG, and C3/A2), the recording montage included the following leads with an average reference to perform spectral analysis: F3, F4, C3, C4, P3, P4, T3, T4, O1, and O2. Recording of oral and nasal airflow, thoracic and abdominal movements, and oxymetry

was performed to rule out apneas and hypopneas. None of the patients had a respiratory disturbance index (apneas and hypopneas) greater than 10 (mean respiratory disturbance index, 2.0 ± 2.8). Surface EMG of the right and left anterior tibialis muscles was recorded to document PLMS. Electrocardiogram was recorded from a standard D1 lead. Microarousals were scored on the C3-A2 EEG lead, using the American Sleep Disorders Association criteria,⁴¹ and the index of microarousals (number per hour of sleep) was calculated.

Electroencephalogram Spectral Analysis

A 16-channel Grass polygraph (amplifier gain $7.5 \mu\text{V/mm}$, bandpass 0.3–100 Hz) was used to amplify the signals, which were relayed to a computer where they were digitized at a sampling rate of 256 Hz. Power spectral analyses by fast Fourier transforms were performed on artifact-free sections (96 seconds in total) of awake and of tonic REM sleep EEGs. The awake EEG was recorded for 10 minutes while subjects were lying in bed with eyes closed in the morning after the night recording. To prevent drowsiness, subjects were asked to open their eyes periodically or when slow rolling eye movements (sign of sleepiness) were noted. The samples of REM sleep were visually selected in the following manner: the first artifact-free sections clearly positioned between two bursts of REMs were visually selected from the first, middle, and last REM sleep periods. EEG sections concurrent with PLMS were carefully excluded from analysis because changes in EEG activity have been described in association with PLMS, namely, an increase in either α or θ and δ power after the onset of PLMS.⁴² A commercial software package (Harmonic 5.1) computed fast Fourier transforms on 4-second miniepochs with a cosine window tapering yielding a spectral resolution of 0.25 Hz.

Absolute EEG power was calculated from each of the 10 leads during both wakefulness and REM sleep. The possibility of interhemispheric asymmetry was assessed by the $(\text{left} - \text{right})/(\text{left} + \text{right})$ ratio for homologous pairs of electrodes. Because no significant asymmetry in total activity was found in any region, homologous leads were pooled together and five cortical regions were considered: frontal, central, parietal, temporal, and occipital. Five frequency bands, namely, δ (0.75–4.0 Hz), θ (4.0–8.0 Hz), α (8.0–13.0 Hz), β_1 (13.0–22.0 Hz), and β_2 (22.0–32.0 Hz), were defined. Because it was observed that, in the awake EEG, between-group differences in absolute power were essentially restricted to two bands, namely, θ and β_2 , with smaller or negligible modifications in the other bands, the ratio of the power in θ over β_2 (TH/BE2) was calculated as a specific index of cortical slowing that could possibly differentiate RBD patients from controls. Finally, the dominant occipital frequency (DOF) during wakefulness was determined for both groups.

Statistical Analyses

Student's *t* tests were performed to evaluate between-group differences in the PSG variables and significance level was set at *p* value less than 0.05. Between-group differences on absolute power in the five frequency bands for the five cortical regions were assessed for each state by a three-way analysis of variance with Group as an independent factor and Band and

Region as repeated measures. Huynh-Feldt corrections for sphericity were applied to all repeated measure analyses. Between-group differences on the TH/BE2 ratio were assessed statistically by a three-way analysis of variance with two independent factors (Group and Gender) and one repeated measure (Region). Post hoc analyses were done using the Tuckey honest significant difference test. A Mann-Whitney *U* test was performed to assess between-group differences in the DOF during wakefulness. Before statistical analyses, a log transformation was done on power values in the five frequency bands and on the TH/BE2 ratio to normalize data distribution. Finally, the relationship between the EEG slowing, as measured by the TH/BE2 ratio, and age, clinical (symptoms frequency and intensity, disease duration) and polysomnographic (percentage of atonia and phasic EMG activity) indices of RBD severity was assessed by Pearson product moment correlation tests.

Results

Sleep Architecture

Polysomnographic variables are shown in the Table. No between-group difference was found on sleep architecture, namely, sleep latency, total sleep time, sleep efficacy and percentages of stage 1, stage 2, slow-wave sleep, and REM sleep. The two groups did not statistically differ on MA index either.

Spectral Analysis in Wakefulness

ELECTROENCEPHALOGRAPH POWER. A significant Group by Band by Region interaction ($F[16,448] = 2.19$; $p = 0.024$; $\epsilon = 0.55$) was found for EEG power in wakefulness. Simple interaction effects were all significant at p value less than 0.02 for every region. Simple main effects showed that idiopathic RBD patients had a higher θ power in frontal ($p = 0.005$), temporal, ($p = 0.001$), and occipital ($p = 0.024$) regions compared with controls. A similar trend was observed in central ($p = 0.06$) and in parietal ($p = 0.1$) regions.

Table. Polysomnographic Characteristics of Patients with RBD and Control Subjects (mean $\pm$ SEM)

Characteristic	RBD Patients (n = 15)	Control Subjects (n = 15)	<i>p</i>
Sleep latency (min)	37.2 $\pm$ 11.5	20.0 $\pm$ 3.8	NS
Total sleep time (min)	350.8 $\pm$ 18.7	366.2 $\pm$ 16.3	NS
Sleep efficiency (%)	78.3 $\pm$ 3.1	77.9 $\pm$ 2.9	NS
Stage 1 sleep (%)	13.6 $\pm$ 2.0	15.0 $\pm$ 1.6	NS
Stage 2 sleep (%)	61.5 $\pm$ 2.6	61.9 $\pm$ 1.5	NS
SWS (%)	8.1 $\pm$ 1.6	4.3 $\pm$ 1.2	NS
Stage REM sleep (%)	16.9 $\pm$ 1.9	18.8 $\pm$ 1.4	NS
Microarousal index	10.1 $\pm$ 1.7	12.2 $\pm$ 1.5	NS

RBD = rapid eye movement sleep behavior disorder; NS = not significant; SEM = standard error of mean; SWS = slow-wave sleep.

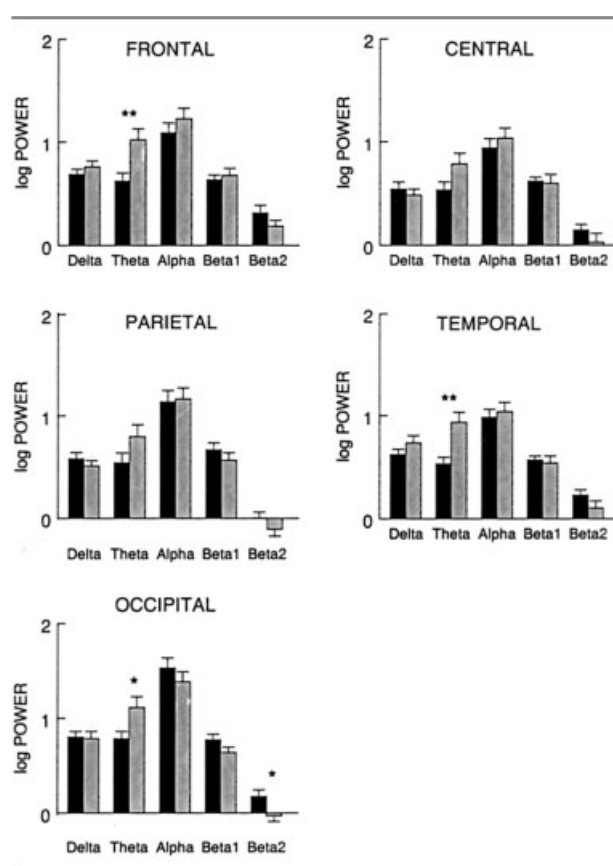


Fig 1. Log-transformed absolute electroencephalogram power in five frequency bands, for five cortical regions, during wakefulness in control subjects (black bars) and in RBD patients (gray bars). * $p < 0.05$; ** $p < 0.01$.

In addition, compared with controls, RBD patients showed a lower spectral power in β_2 ($p = 0.034$) and a trend for a lower β_1 power ($p = 0.1$) in the occipital region. Figure 1 illustrates the log-transformed absolute power of the five bands for each cortical region during wakefulness. In the RBD group, an overall higher θ and a lower β_2 power are observable in every cortical region.

TH/BE2 RATIO. The three-way analysis of variance showed a main effect for Group ($F[1,26] = 9.07$; $p = 0.006$) and for Region, ($F[4,104]$; $p < 0.0001$; $\epsilon = 0.99$), indicating a higher TH/BE2 ratio in the RBD group and an overall higher ratio in the occipital region. A main effect for Gender was also found ($F[1,26] = 6.88$; $p = 0.014$), indicating that men, in general, have higher value of TH/BE2 ratio than women. A strong trend for an interaction between Group and Gender was observed ($p = 0.057$), suggesting that men with RBD have a higher TH/BE2 ratio than that of women with RBD or male or female controls. This is clearly shown in Figure 2.

Three patients presented the highest values of the

TH/BE2 ratio in all regions. However, these three patients did not differ drastically from the other patients on clinical characteristics of the disorder, namely, duration of illness, frequency, or intensity of the clinical symptomatology, REM sleep variables such as percentage of atonia and phasic EMG activity, or age. No correlation was found either in the entire RBD group between these variables or the TH/BE2 ratio.

DOMINANT OCCIPITAL FREQUENCY. The mean DOF during wakefulness was $8.6 \pm 1.0\text{Hz}$ in the RBD patients and $9.2 \pm 0.7\text{Hz}$ in normal controls; this between-group difference was significant ($U = 63.5$; $p = 0.042$). Moreover, when only men were considered, the mean DOF of RBD patients was $8.3 \pm 0.9\text{Hz}$, 1Hz below the value of normal controls ($9.3 \pm 0.6\text{Hz}$; $U = 18.5$; $p = 0.017$). Of interest, four idiopathic RBD male patients, including the three identified as having the highest TH/BE2 ratio, presented a DOF value below 8Hz, thus in the θ range. Nonetheless, the value of the DOF was reduced in most RBD patients. In fact, as shown in Figure 3, the mean power spectrum of the RBD group appears to be shifted toward slower frequencies compared with that of the control group.

Spectral Analysis in Rapid Eye Movement Sleep

A significant Group by Band by Region interaction ($F[16,448] = 3.05$, $p < 0.0001$, $\epsilon = 0.68$) was found for REM sleep. Simple interaction effects (Group by Region) were significant at p value less than 0.03 for all

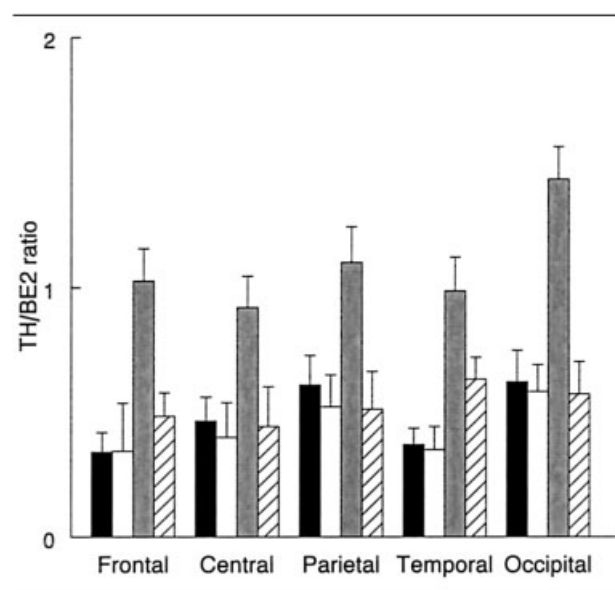


Fig 2. Log-transformed values of TH/BE2 ratio for five cortical regions, during wakefulness, in male control subjects (black bars), female control subjects (white bars), male rapid eye movement sleep behavior disorder (RBD) patients (gray bars), and female RBD patients (hatched bars).

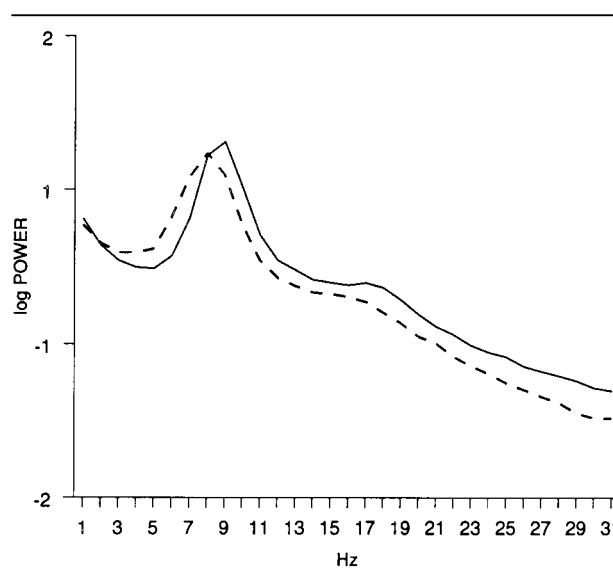


Fig 3. Log-transformed power spectrum in the occipital region for control subjects (continuous line) and rapid eye movement sleep behavior disorder patients (dotted line).

bands except θ . Simple main effects showed that, compared with controls, idiopathic RBD patients had a lower spectral power in $\beta 1$ ($p = 0.044$) and a similar trend in $\beta 2$ ($p = 0.068$), both in the occipital region, with no between-group difference for the remaining bands.

Discussion

RBD traditionally has been considered as a parasomnia caused by a motor dysfunction occurring only during REM sleep. The results of this study show, however, that patients with idiopathic RBD may also present an impairment of cortical activity during both REM sleep (lower β power) and wakefulness (lower β and higher θ power).

One may argue that the elevated θ power observed in RBD could be a consequence of the parasomnia-related sleep disruption, because an increase of θ power (associated with an increase of β power) has been demonstrated in normal healthy subjects undergoing sleep deprivation.^{43–45} However, because no between-group differences in sleep architecture variables, including total sleep time, sleep efficacy, and MA index, were observed during prior nocturnal PSG and a lower (rather than higher) β power was obtained in the RBD group, the hypothesis of sleep deprivation seems unlikely.

Changes in waking EEG activity reported here in RBD are similar to those observed in association with Alzheimer's disease (AD)^{46,47} and, to a lesser degree, in normal aging.⁴⁸ In particular, an increase of θ activity has been reported in the initial stages of dementia, representing a sensitive index of early cognitive deterioration.^{49,50} Slowing of the dominant occipital α rhythm

is also observed in several neurodegenerative diseases, as AD,^{51,52} PD,^{53,54} and DLB.^{55,56} On the other hand, fast activities appear to be correlated with cognitive performance in normal elderly population,⁵⁷ and isolated decrease of β activity was indeed found to be associated with early signs of intellectual loss.⁵⁸ Similarly, the diffuse slowing of the EEG observed in full-blown neurodegenerative diseases such as AD,^{59–61} DLB,⁵⁵ PSP,⁶² and PD with dementia^{53,54} appears to be closely related to cognitive deterioration. In AD, this diffuse EEG slowing had been also shown to occur in both wakefulness and REM sleep.⁶³

As mentioned, RBD often is associated with, but may precede by many years,^{7,12–16} neurodegenerative disorders featuring a cognitive impairment. However, very few data concerning the long-term evolution of idiopathic RBD are available. It may be postulated that the EEG slowing observed in idiopathic RBD patients, and expressed by an increase of TH/BE2 ratio, would represent a very early sign of a central nervous system dysfunction, perhaps paralleled by subtle cognitive losses, that eventually could evolve toward a more severe impairment.

Cognitive functions appeared to be preserved in our RBD population, as assessed by a standard clinical evaluation. Note that none of the patients (or their relatives for them) reported any sign of cognitive decline at the time of the study. In addition, six male patients, including two of the three patients with the more severely affected EEG, underwent a Mini-Mental State Examination,⁶⁴ which was in all cases higher than 28 (mean score, 29.16 ± 0.89). However, two RBD patients with normal Mini-Mental State Examination score underwent an extensive neuropsychological evaluation that showed an impairment in visuospatial planning and a deficit in the recall of both a visuoconstructive task and verbal material. These results are in agreement with a previous neuropsychological study performed in RBD patients of nonspecified cause.⁶⁵ Similar deficits, although more severe, are observed in demented patients, including those with DLB^{66,67} and PD,⁶⁸ as well as in PD patients without dementia.⁶⁹ Therefore, neuropsychological studies of idiopathic RBD patients that include a detailed assessment of cognitive functions known to be impaired in the aforementioned disorders are particularly needed. In addition, our results suggest that eventual correlation studies between cognitive and EEG measures should be performed for men and women separately.

The idiopathic RBD patients showing the highest values of TH/BE2 ratio may represent a distinct subgroup of patients who are more likely to eventually develop a degenerative dementia. Alternatively, the distribution of the TH/BE2 ratio values may reflect a neurodegenerative process unique to RBD at different stages of evolution. The lack of correlation between the

ratio values and disease duration and clinical severity speaks more in favor of the first hypothesis than of the second. One has to keep in mind that these subjective variables are difficult to establish with any precision. However, the objective measures such as percentages of muscle atonia or of phasic EMG activity during REM sleep also failed to correlate with EEG slowing. A longitudinal assessment of the EEG and cognitive and neurological profile in idiopathic RBD will help to clarify this issue.

The existence of a striking male predominance in RBD is well known.³ This study shows that RBD also seems to affect the cortical activity much more severely in men than in women. We cannot offer a proper explanation for this observation, except to relate this phenomenon to the gender differences frequently observed in EEG during wakefulness⁷⁰ and sleep.⁷¹ Whether women with RBD would be less likely than men to eventually develop a degenerative dementia still remains to be ascertained.

The impairment of cortical activity observed in idiopathic RBD patients was particularly evident in the occipital region during both wakefulness and REM sleep. An impairment of frontal and temporal regions also was observed only during wakefulness. Interestingly, this topographical distribution of EEG changes is reminiscent of the perfusion and metabolic impairment pattern observed in DLB^{72–74} and, to some extent, in PD.^{75–77}

The pathogenesis of idiopathic RBD has been variously attributed to a dysfunction of the LC, the PPN region, or nigrostriatal systems. Some of the main EEG features in RBD, such as the increase in θ power and the shift of the power spectrum toward slower frequencies in the occipital region, were found only during wakefulness. The noradrenergic neurons of LC, maximally discharging during wakefulness and virtually silent during REM sleep,⁷⁸ are known to be implicated in the modulation of cortical activation during wakefulness,³¹ through their widespread projections to the forebrain and cortex.²⁷ Their impairment could contribute to the increase of θ power observed during wakefulness in RBD patients. Midbrain dopaminergic neurons originating from both the substantia nigra and the ventral tegmental area, through projections to the neostriatum, the basal forebrain, and the frontal cortex, are thought to be implicated in behavioral arousal and alertness mechanisms.²⁷ Thus, a nigrostriatal dopaminergic impairment also could play a role in the EEG slowing observed during wakefulness in RBD patients. Finally, the decrease of β power during both wakefulness and REM sleep might be ascribed to a dysfunction of the PPN region, because this cholinergic system appears to be involved in EEG activation in both wakefulness and REM sleep.^{26,79} The pathophysiological

mechanism of the impaired cortical activation in idiopathic RBD patients deserves further investigation.

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